Question:

A 34-year-old woman comes to the emergency department complaining of severe headache and anxiety, diaphoresis, and palpitations for the last 30 minutes. She has had several similar episodes over the past few weeks. She has no significant past medical history and has a 10 pack-year smoking history. She takes no illicit drugs. Her blood pressure on arrival is 181/80 mmHg and her pulse is 134/min. If this patient was given metoprolol, how would her blood pressure respond?Answer Choices:A. Hypotension due to beta-adrenergic receptor blockadeB. Hypotension due to alpha-1-adrenergic receptor blockadeC. Hypertension due to alpha-1-adrenergic receptor stimulationD. Hypertension due to alpha- and beta-adrenergic receptor blockade

Reasoning + Answer:

<think> The patient is presenting with symptoms of acute anxiety attack, which is likely caused by a catecholamine surge. Metoprolol is a beta-blocker that selectively blocks beta-1 and beta-2 adrenergic receptors, reducing the effects of catecholamines on the heart and peripheral arteries. By blocking beta-2 receptors, metoprolol causes peripheral vasodilation, which can lead to a decrease in blood pressure. The patient's blood pressure is already elevated at 181/80 mmHg, and the administration of metoprolol would likely further decrease her blood pressure due to the vasodilatory effects. The other options are incorrect because metoprolol does not significantly block alpha-1 adrenergic receptors, and its blockade of these receptors is not the primary mechanism by which it lowers blood pressure. Additionally, metoprolol would not cause hypertension due to receptor stimulation. </think><answer> A. Hypotension due to beta-adrenergic receptor blockade</answer>